# Project Questions

Design and Analysis of Algorithms

Instructions:

- Design an efficient algorithm for the selected question and implement it in a programming language of your choice.
- Prepare a project report containing a brief description of the algorithm and a detailed theoretical analysis. Include a link to your implementation, along with a short README explaining how to run the code.
- Submit the project report as a PDF. Use a typesetting language such as LaTeX to prepare the document.
- Marks will be based not only on the correctness of your algorithm but also on the clarity of your writing — submit a clear, well-organized, and formal document.
- A viva will be conducted for this project. Be prepared to present and explain your code and walk through test cases.
- This project must be completed independently, without external help.

## Questions

**Common text for Questions 1 to 5 :** In organization ABC, the employees have an underlying hierarchy, where the CEO is at the top and every other employee has exactly one manager. An employee can be the manager to one or more employees. Any employee is familiar with the work of her manager and the works of her immediate sub-ordinates.

1. It was decided to give away gift coupons for all employees(including the CEO) as New year presents. By mistake, the gift coupons ordered were of three different denominations - $1000, 2000$ and $3000$. It was decided that every employee will receive a gift coupon of denomination different from her manager. However, if an employee receives a gift

coupon of larger denomination than her manager, she is sure to be given a negative rating by her manager in a fit of jealousy. Design an algorithm to decide how to distribute gift coupons among employees so that the number of employees who gets a negative rating is minimized.

2. ABC decides to constitute a communication team. A communication team consists of employees such that for every employee who is not in the team, there is a person in the communication team that is familiar with her work. Further the management decides that for every employee who is not in the team, there is exactly one person in the communication team that is familiar with her work. Give an algorithm that selects such a communication team of the smallest possible size.

3. ABC decides to constitute a communication team. A communication team consists of employees such that for every employee who is not in the team, there is a person in the communication team that is familiar with her work. Further the management decides that if the team consists of a particular employee, then her manager cannot be present in the communication team. Give an algorithm that selects such a communication team of the smallest possible size or return such a team cannot exist.

4. ABC decides to constitute a communication team. A communication team consists of employees such that for every employee who is not in the team, there is a person in the communication team that is familiar with her work. Further the management decides that if the team consists of a particular employee, then her manager cannot be present in the communication team and for every employee who is not in the team, there is exactly one person in the communication team that is familiar with her work. Give an algorithm that selects such a communication team of the smallest possible size. or return such a team cannot exist.

5. ABC further decided to have a restricted communication team with the constraint that for every employee, there is some one in the communication team who is at most 2 levels away in the hierarchy. i.e, for every employee, either she is in the communication team OR her manager is in the communication team OR her manager's manager is in the communication team OR one of her immediate subordinates is in the communication team OR one of the immediate subordinate of her immediate subordinate is in the communication team. Design an algorithm to select such a communication team of the minimum possible

size.

6. One day, Alex got tired of climbing in a gym and decided to take a very large group of climber friends outside to climb. The climbing area where they went, had a huge wide boulder, not very tall, with various marked hand and foot holds. Alex quickly determined an "allowed" set of moves that her group of friends can perform to get from one hold to another. The overall system of holds can be described by a rooted tree T with n vertices, where each vertex corresponds to a hold and each edge corresponds to an allowed move between holds. The climbing paths converge as they go up the boulder, leading to a unique hold at the summit, represented by the root of T. Alex and her friends (who are all excellent climbers) decided to play a game, where as many climbers as possible are simultaneously on the boulder and each climber needs to perform a sequence of exactly k moves. Each climber can choose an arbitrary hold to start from, and all moves must move away from the ground. Thus, each climber traces out a path of k edges in the tree T, all directed toward the root. However, no two climbers are allowed to touch the same hold; the paths followed by different climbers cannot intersect at all. Describe and analyze an efficient greedy algorithm to compute the maximum number of climbers that can play this game.

7. One day, Alex got tired of climbing in a gym and decided to take a very large group of climber friends outside to climb. The climbing area where they went, had a huge wide boulder, not very tall, with various marked hand and foot holds. Alex quickly determined an "allowed" set of moves that her group of friends can perform to get from one hold to another. The overall system of holds can be described by a rooted tree T with n vertices, where each vertex corresponds to a hold and each edge corresponds to an allowed move between holds. The climbing paths converge as they go up the boulder, leading to a unique hold at the summit, represented by the root of T. Alex and her friends (who are all excellent climbers) decided to play a game, where as many climbers as possible are simultaneously on the boulder and each climber needs to perform a sequence of exactly k moves. Each climber can choose an arbitrary hold to start from, and all moves must move away from the ground. Thus, each climber traces out a path of k edges in the tree T, all directed toward the root. However, no two climbers are allowed to touch the same hold; the paths followed by different climbers cannot intersect at all. Describe and analyze an efficient dynamic programming algorithm to compute the maximum number of climbers that can play

this game.

8. There are $m$ houses in a street. It is needed to build transmitters such that for every house in the street, there should exist a transmitter within $500m$ of it. There are $n$ possible locations where transmitters can be built. Also, in order to avoid signal interference, every house should have *exactly one* transmitter within $500m$ of it. Show that there may be instances where a solution does not exist at all, by giving such an example. Give an algorithm to find the optimal locations for transmitters such that minimum number of transmitters are built , if a solution exists, or correctly conclude that a solution does not exist.

9. There are $m$ houses in a street. It is needed to build transmitters such that for every house in the street, there should exist a transmitter within $500m$ of it. There are $n$ possible locations where transmitters can be built. Also, in order to avoid signal interference, it is recommended that there exists *exactly one* transmitter within $500m$ of a house. Give an algorithm to find the optimal locations for transmitters such that the number of houses that has more than one transmitters within $500m$ of it is minimized.

10. There is a circular road around a military building complex and there are $n$ check posts built around the perimeter of the circular road. For efficient communication between check posts, it is decided to built transmitters along the road so that every check-post has a transmitter within 250m of it. (The distance is measured as the length of the shortest arc between two points along the perimeter of the circle) $m$ possible locations are identified where transmitters can be built. However, in order to avoid signal interference, it is recommended that there exists *exactly one* transmitter within $250m$ of a check post. Give an algorithm to find the optimal locations for transmitters such that the number of check-posts that has more than one transmitters within $250m$ of it is minimized.

11. There is a circular road around a military building complex and there are $n$ check posts built around the perimeter of the circular road. For efficient communication between check posts, it is decided to built transmitters along the road so that every check -post has a transmitter within 250m of it. (The distance is measured as the length of the shortest arc between two points along the perimeter of the circle) . $m$ possible locations are identified where transmitters can be built. However, in order to avoid signal interference, it is recommended that there

exists *exactly one* transmitter within $250m$ of a check post. Give an algorithm that finds the optimal locations for transmitters such that no check-post has more than one transmitter within $250m$ of it or returns that such a solution is not possible.

**Common Text for Questions 12-14 :** Country XYZ has $n$ important towns, which are connected by roads. There are only $n-1$ roads and the road connectivity is such that every town is reachable from every other town by a unique sequence of roads. Two towns are said to be *neighbors* if there is a direct road connecting them. The king of country XYZ decided to station military troupes in the towns to counter possible attacks from enemies. Since the country lacks sufficient military budget, the king decided not to post a troupe in every town.

12. Every town has the option of posting $0, 1$ or 2 troupes such that any town with 0 troupes has at least one neighboring town with 2 troupes. Design an algorithm to assign troupes to towns so that the total number of troupes assigned is minimized.

13. Every town has the option of posting $0, 1$ or 2 troupes such that any town with 0 troupes has at least one neighbor with 2 troupes or has at least two neighboring towns with 1 troupe. Design an algorithm to assign troupes to towns so that the total number of troupes assigned is minimized.

14. Every town has the option of posting $0, 1$ or 2 troupes such that any town with 0 troupes has exactly one neighboring town with 2 troupes. Design an algorithm to assign troupes to towns so that the total number of troupes assigned is minimized.

**Common text for Questions 15 - 18 :** Lectures in an educational institution runs around the clock. There are $n$ lectures and the schedule of the lectures is the same every day. The duration of lectures may vary from course to course. There are some lectures which spans across different days, for example, starting at 11 PM and ends at 1 AM the next day. Hence the earliest lecture or last lecture is not well defined.

15. The supervisor of the institution wants to check on every lecture in a day. For this, she makes a series of visits such that at the time of a visit, she checks on every lecture happening at that instant. In a day, there are $m$ discrete times when the supervisor is free for visits. Give an algorithm that outputs the minimum number of visits the supervisor has to make so as to check on every lecture happening in a day.

16. The supervisor of the institution wants to check on every lecture in a day. For this, she makes a series of visits such that at the time of a visit, she checks on every lecture happening at that instant. In a day, there are $m$ discrete times when the supervisor is free for visits. Also, in order to minimally disturb the lectures, it is advised that every lecture is checked exactly once. Give an algorithm that outputs the minimum number of visits the supervisor has to make so as to check on every lecture happening in a day exactly once, if that is possible.

17. The supervisor decided to select some courses such that the TAs of these courses can supervise the lectures that overlaps with the lecture of that course(Assume that any course conducts at most one lecture in a day). Now the problem boils down to selecting a subset of lectures such that every lecture overlaps with at least one lecture from this subset. Moreover, to ensure minimal disturbance it is to be ensured that every lecture overlaps with exactly one lecture from this subset(including itself).Design an algorithm to select such a subset.

18. The supervisor decided to select some courses such that the TAs of these courses can supervise the lectures that overlaps with the lecture of that course(Assume that any course conducts at most one lecture in a day). However TAs of a course cannot supervise an lecture of the same course. Now the problem boils down to selecting a subset of lectures such that every lecture overlaps with at least one lecture from this subset. Moreover, to ensure minimal disturbance it is to be ensured that every lecture overlaps with exactly one lecture from this subset(excluding itself).Design an algorithm to select such a subset.

19. During lock down, a parent is working from home with a toddler who has free time from school. The parent has noticed that the toddler gets cranky at specific times every day( for example, at 9 AM, 10.45 AM, 12.15 PM etc) and decides to enroll her in some online activities during this time. On contacting an online activity center for kids, she is given a schedule of activities that happens between 6 AM and 10 PM and every activity has an enrollment fee. (For example, story telling session from 8.30 AM to 9.30 AM with a fee of Rs. 2000, puppet show from 11 AM to 11.30 AM with a fee of Rs. 1500 etc.). Design an algorithm that helps the parent to decide on a schedule of activities such that the kid is enrolled in some activity during any time she feels cranky, no two scheduled activities overlap and the total enrollment fees is minimum possible.

20. Given a set of points $P$ arranged along the boundary of a circle and a family of arcs $\mathcal{A}$ such that every arc in $\mathcal{A}$ contains a consecutive set of points from $P$. Find a subset $A' \subseteq \mathcal{A}$ such that every point in $P$ is contained in at least one arc of $A'$ and the number of points contained in exactly one arc of $A'$ is maximised.